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# Measuring ε in the brain's own units

*Detectability-calibrated adversarial training for CEBRA neural decoders — what we built, what failed, and what holds across 378 trained models*

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EPFL · Experiments 1–8 · Perich motor-cortex recordings (9 sessions, 4 monkeys)

## Abstract

Adversarial training of neural-population encoders inherits a perturbation budget designed for images: a radius ε in raw input units, identical in every direction. For spike counts this number has no physical meaning, it is not comparable between recordings and, as we show, it leaves decoders open to perturbations that look exactly like the recording's own noise. Following our research note, we measure a perturbation by *how detectable it is against that noise*: ε becomes the ideal-observer sensitivity d′ under a space×time (Kronecker) noise covariance, so that $\varepsilon = 1$ is one just-noticeable difference (JND). We train CEBRA encoders with d′-constrained APGD and evaluate every model in two shared threat metrics. Across 9 sessions, 3 seeds and both CEBRA modes (378 encoder–decoder pairs in the final comparison), noise-aware training raises the noise-shaped robustness threshold 1.2–1.3× above tuned L2-PGD (27/27 paired runs in time-only CEBRA) without loss of clean decoding, at the price of a quantifiable trade-off against isotropic attacks. Two variants that follow the note — label-free *difference* residuals and a covariance *pool* — stay robust in both CEBRA modes, and the pool is the first variant that also improves clean $R^2$ over tuned PGD (+0.022, 9/9 sessions, 4/4 animals).

## Key findings

1. **Clean decoders break below one JND.** Under noise-shaped perturbations, every clean time-only CEBRA decoder loses half of its $R^2$ at $d' \leq 0.73$ (0/27 runs survive $d' = 1$), while a blind anomaly detector stays at chance up to $d' = 4$ in all nine sessions.
2. **Tuned L2-PGD only partly closes the gap** (half-damage threshold $d' \approx 1.2$; 18/27 runs survive one JND), and its best raw ε differs between sessions (3, 6 or 12).
3. **Noise-aware d′ training** raises the noise-shaped threshold by 1.29× (SP8), 1.25× (DIFF8) and 1.21× (POOL) relative to tuned L2-PGD (PGD6, $\varepsilon = 6$) in **27/27** time-only runs (9/9 sessions, sign test $p = 0.004$), with clean $R^2$ unchanged; DIFF8 survives one JND in 27/27 runs.
4. **The price is isotropic robustness** (0.49–0.82×). At equal damage, the isotropic attacks that still work against noise-aware models are readily detected in low-firing sessions (CNN AUC up to 0.98; $\rho \approx -0.7$ with firing rate), and in d′ units the cheapest attack is always the noise-shaped one.
5. **Behaviour-supervised CEBRA:** DIFF8 (1.32×, 9/9 sessions) and POOL (1.20×, 8/9) keep the gain; the whole-covariance SP8 over-shoots and becomes brittle in 5/9 sessions. The noise model must describe *noise*, not the slow behavioural signal.

| Recordings | Monkeys | Seeds | CEBRA modes | Models (final) | Training runs |
|---|---|---|---|---|---|
| 9 | 4 | 3 | time-only + supervised | 378 | ≈ 530 (Exp. 1–8) |

# 1 The story in one page

**Why ε matters.** ACORN showed that adversarial training (AT) markedly improves time-contrastive CEBRA embeddings of motor cortex — in our nine sessions, clean decoding $R^2$ rises by +0.14 over standard CEBRA. But AT needs a perturbation budget, and the one we inherited comes from computer vision: an L2 (or L∞) ball in raw input units. A pixel unit has a fixed perceptual meaning; a spike-count unit does not. One spike per bin is negligible for a neuron firing at 50 Hz and enormous for a neuron that is almost silent, and a 36-bin window of 38 neurons has a very different natural variability from one of 145. A raw-unit ε is therefore an arbitrary, session-specific hyperparameter, and an isotropic ball spends the same budget on implausible directions as on natural ones.

**The idea (from the research note).** Measure a perturbation by how detectable it is against the recording's own noise. For Gaussian noise with covariance Σ the optimal detector of a known perturbation δ has sensitivity $d' = (\delta^\top \Sigma^{-1} \delta)^{1/2}$. Using $d'$ as the norm makes ε dimensionless and interpretable: $\varepsilon = 1$ is one just-noticeable difference, at which an observer who knows δ *exactly* is correct only 69% of the time. The same metric also fixes the attack direction — steepest ascent under it is $\Sigma \nabla_x L$, i.e. along the directions in which the data naturally vary.

**What we did.** We implemented the note completely, tested it on one recording, then pushed it through increasingly demanding comparisons. The table below is the honest path, including what did not work.

**Table 1.** The experimental path. Exp. 7 (external dataset) is planned and not part of this report.

| Exp. | Design | What we learned |
|---|---|---|
| 1–2 | C-CO12, 24 + 4 configs: all three layers of the note (stationary metric, trial-level masking, physiological guards) and the three-way ε* calibration | Guards and trial masks do not help on spike counts. Calibration is self-consistent: ε* = 1, and learned detectors are at chance at $d' = 1$. |
| 3 | Madry vs TRADES at $d' \in \{0, 1, 3, 8\}$, 3 seeds | Near-zero-ε controls rule out a dropout-consistency artefact (TRADES $d'1 - d'0 = +0.10$ $R^2$); clean $R^2$ rises with the budget up to $d' \approx 8$. |
| 4 | 12 arms incl. isotropic and spatial-only controls; first held-out robustness evaluation | Equal nominal $d' \neq$ equal perturbation: the noise-aware ball is ≈1.8× larger in raw units, so geometries must be compared at matched size or at their optima. |
| 5 | 9 sessions × 12 arms (seed 42) | Noise-shaped robustness gain in 9/9 sessions; clean $R^2$ ties; automatic budget from blind detectability. |
| 6 | 3 seeds × 6 arms (time-only); equal-damage detectability of 108 saved models | Gain replicates 27/27; covariance pool improves clean $R^2$; isotropic trade-off is concentrated in detectable attacks. |
| 8 | Behaviour-supervised CEBRA, 3 seeds × 8 arms (216 models) | DIFF8 and POOL keep the gain; whole-covariance SP8 over-shoots — the noise model must describe noise. |

# 2 Why ordinary PGD is not enough for neural data

## 2.1 A raw radius has no physical meaning

Write the noise covariance as $\Sigma = \sum_i \lambda_i u_i u_i^\top$. A perturbation of L2 size r along the eigen-direction $u_i$ has detectability

$$d'_i = r/\sqrt{\lambda_i}, \qquad \max_i d'_i / \min_i d'_i = \sqrt{\kappa(\Sigma)} \tag{1}$$

In our recordings the spatial condition number κ of the noise covariance ranges from 22 to 4,905 (largest in low-firing sessions with near-silent neurons), so the *same* raw radius is 5–70× more detectable along the quietest direction than along the dominant one. The isotropic ball wastes budget on directions that never occur naturally and barely touches the ones that do (Figure 1).

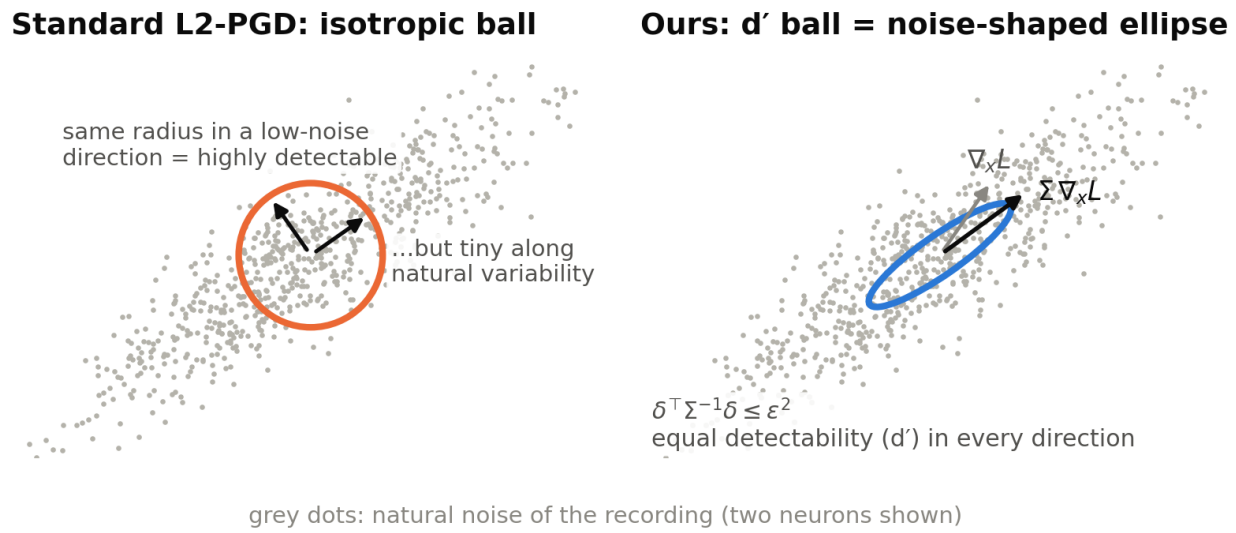



**Figure 1.** Isotropic versus noise-shaped budgets on a two-neuron toy example. Left: an L2 ball has the same radius in every direction, which is a tiny change along the main axis of natural variability and a large, unnatural change across it. Right: the d′ ball $\delta^\top \Sigma^{-1} \delta \le \varepsilon^2$ is an ellipse aligned with the noise; its steepest-ascent direction is the noise-coloured gradient $\Sigma \nabla_x L$.

## 2.2 It is a per-session hyperparameter

With raw-unit L2-PGD the best ε (of 3, 6 and 12) was 6 in five sessions, 3 in three and 12 in one (Exp. 5). Too large a raw ε is costly: ε = 12 lowers clean R² by 0.085 on average relative to ε = 6.

## 2.3 Decoders break below detectability

We attack the full encoder + decoder on held-out data with APGD constrained to the noise-shaped d′ ball (Section 3.6). Clean CEBRA decoders lose half of their R² at d′ ≤ 0.73 in all 27 time-only runs; tuned L2-PGD raises this threshold only to a median d′ of 1.19. At these budgets a blind autoencoder detector — an observer that does not know the attack — is at chance (AUC ≤ 0.508 up to d′ = 4 in every session; Figure 2b). In other words, the conventional defence leaves decoders breakable by perturbations that cannot be noticed without knowing the attack.

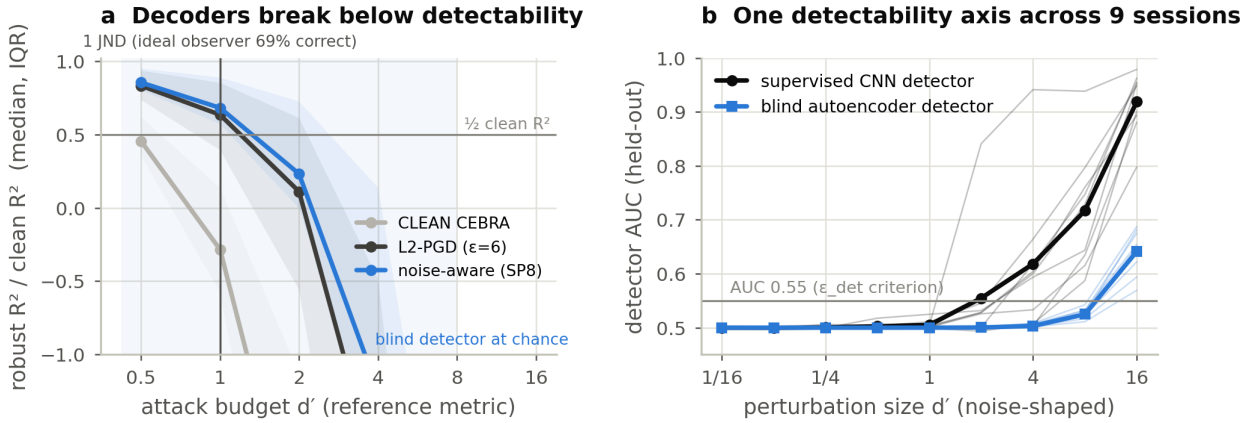



**Figure 2.** (a) Robust R² relative to clean R² under the noise-shaped attack, time-only CEBRA (median and IQR over 27 runs: 9 sessions × 3 seeds). The vertical line marks one JND (d′ = 1); the shaded band is the range in which the blind detector is at chance. (b) Held-out detector AUC versus perturbation size for the nine sessions (thin lines) and their mean (thick): the supervised CNN knows the attack distribution, the autoencoder only sees clean data. Both curves are nearly identical across sessions — d′ is a common axis.

### 2.4 Robustness is geometry-specific — and size does not buy it

Training against one geometry protects against that geometry. A larger isotropic budget does not substitute for the right shape: the isotropic ISO16 model trained with perturbations that were larger than those of the noise-aware NA4 model by every measure (median raw L2 9.0 vs 4.6, reference d′ 10.0 vs 4.0) is nevertheless less robust to noise-shaped attacks (half-damage threshold 1.19 vs 1.77, 9-session means). What a model becomes robust to is set by the geometry of its training perturbations.

## 3 Method: ε as a detectability index

### 3.1 Detectability and the just-noticeable difference

For additive Gaussian noise with covariance Σ, the ideal observer that knows the perturbation δ detects it with sensitivity d′ and proportion correct (unbiased yes/no task)

$$d'(\delta) = \sqrt{\delta^\top \Sigma^{-1} \delta}\,, \qquad P_{\text{correct}} = \Phi(d'/2) \tag{2}$$

We use d′ as the attack norm, so the budget ε is a number of noise standard deviations along the perturbation. $\varepsilon_{\text{JND}} = 1$ gives $\Phi(0.5) = 0.69$; $d' = 2, 4, 8$ give 0.84, 0.98 and >0.999.

### 3.2 A noise model that is cheap to whiten

For windows of C channels × T bins we model the covariance as a Kronecker product of a spatial and a temporal factor, estimated on noise residuals R of the training recording:

$$\Sigma = \Sigma_s \otimes \Sigma_t\,, \qquad \Sigma_s = \text{LedoitWolf}(R)\,, \qquad \Sigma_t = \text{Toeplitz}(\text{Welch}(R)) \text{ or } I \tag{3}$$

Whitening a window costs $O(C^2T + CT^2)$. The spatial-only variants used in the final experiments set $\Sigma_t = I$ (white temporal factor); the full Kronecker model (NA) is included in Exp. 5. The temporal factor matters less than expected: the residual lag-1 correlation is ≈ 0.15, and with residuals that contain slow behavioural signal it can steer attacks onto that signal.

### 3.3 What counts as noise: three residual definitions

$$R_t^{\text{whole}} = x_t - \bar{x}\,, \qquad R_t^{\text{diff}} = \frac{x_t - x_{t-1}}{\sqrt{2}}\,, \qquad R^{\text{reg}} = X - Y\hat{B} \tag{4}$$

*Whole* residuals keep everything, including the slow, high-variance population modes that carry behaviour. *Regression* residuals remove what is linearly predictable from the six behavioural variables (available only when the encoder itself is supervised). The label-free *difference* residual is our addition: if $x_t = s_t + n_t$ with a slow signal s and temporally white noise n,

$$\text{Cov}\left[\frac{x_t - x_{t-1}}{\sqrt{2}}\right] = \Sigma_n + \frac{1}{2}\,\text{Cov}[s_t - s_{t-1}] \approx \Sigma_n \tag{5}$$

so the difference residual estimates the covariance of the fast fluctuations — the noise — rather than of the whole activity.

### 3.4 d′-PGD and adversarial training

The attack runs ordinary APGD in whitened coordinates z with an L2 ball of radius ε and maps back:

$$\delta = \Sigma^{1/2} z, \qquad \delta^\star = \Sigma^{1/2} \arg\max_{\|z\|_2 \le \varepsilon} \mathcal{L}(x + \Sigma^{1/2} z), \qquad \nabla_z \mathcal{L} = \Sigma^{1/2} \nabla_x \mathcal{L} \tag{6}$$

In input space every step therefore moves along $\Sigma\nabla_x L$. Training follows the ACORN (Madry) structure — a clean update followed by an update on the adversarial reference sample of the InfoNCE loss:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{InfoNCE}}(x), \qquad \theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{InfoNCE}}(x + \delta^\star(x)) \tag{7}$$

The **pool** variant implements the note's suggestion to "sample Σ from a pool so robustness is not tied to one noise structure": at every step the spatial factor is drawn from the recording's own Ledoit–Wolf estimate or its isotropic counterpart,

$$\Sigma_s^{(k)} \sim \text{Uniform}\{\Sigma_{\text{LW}},\ \bar{\sigma}^2 I\}, \qquad \bar{\sigma}^2 = \frac{1}{C} \operatorname{tr} \text{Cov}(R) \tag{8}$$

### 3.5 Calibration: three budgets that the data define

**(i) Ideal observer:** $\varepsilon_{\text{JND}} = 1$. **(ii) Natural variability:** two independent noise draws of the same window differ by

$$\|\Sigma^{-1/2}(n_1 - n_2)\|_2 \approx \sqrt{2D}, \qquad D = CT \tag{9}$$

and the measured nearest-neighbour distances ($\varepsilon_{\text{NN}}$ = 39–86) sit at 0.75–0.89 × $\sqrt{(2D)}$: in high-dimensional windows natural repeats are separated mainly by noise, so a per-sample $\varepsilon_{\text{NN}}$ bound can never bind. **(iii) Blind detectability:** an observer who does not know δ can at best test the whitened energy, whose mean shifts by $\|\delta\|^2_\Sigma$ against a standard deviation of $\sqrt{(2D)}$ — the square-root law of covert communication:

$$d'_E = \frac{\|\delta\|_\Sigma^2}{\sqrt{2D}} \qquad \Longrightarrow \qquad \varepsilon_{\text{blind}} \approx (2D)^{1/4} \tag{10}$$

Empirically, the blind autoencoder detector crosses AUC 0.55 at d′ = 8.3–12.7, against $(2D)^{1/4}$ = 7.2–10.1 (mean ratio 1.13). The supervised CNN detector crosses it earlier (d′ = 1–4). We use $\varepsilon_{\text{JND}} = 1$ as the threat budget of the evaluation and report robustness relative to it.

### 3.6 Evaluation protocol

Every model — clean, raw-L2 or d′-trained — is evaluated in the same two metrics, both fitted on the training recording: the **reference** (noise-shaped) metric Ledoit–Wolf ⊗ Welch on whole residuals, and the **isotropic** metric $d'_{\text{iso}}(\delta) = \|\delta\|_2 / \bar{\sigma}$ (an L2 ball in units of the mean noise SD). The attacker maximises the standardised squared decoding error of encoder + decoder with 20-step APGD, warm-started along ε ∈ {0.5, 1, 2, 4, 8, 16}, keeping each window's worst perturbation over all budgets ≤ ε (a monotone worst-case curve). Our main summary is the half-damage threshold

$$\varepsilon_{1/2} = \min\left\{ \varepsilon :\ R^2_{\text{adv}}(\varepsilon) \le \frac{1}{2} R^2_{\text{clean}} \right\}, \qquad \text{JND margin} = \varepsilon_{1/2} / \varepsilon_{\text{JND}} \tag{11}$$

A JND margin above 1 means the decoder keeps more than half of its $R^2$ against every perturbation that is at most one just-noticeable difference. Minimum-norm (DDN) margins, contrastive margins and one draw of natural noise complete each evaluation; for detectability, the attacks at $\varepsilon_{1/2}$ of each geometry are scored by a blind autoencoder and by a CNN trained on the attack (Section 5.5).

# 4 Experimental setup

**Table 2.** Data and protocol.

| Component | Configuration |
|---|---|
| Data | Perich motor-cortex recordings, raw spike counts; sessions C-CO12, C-CO13, C-RT13, C-RT6 (monkey C), M-CO9, M-RT3 (M), T-RT1, T-RT4 (T), J-CO0 (J); 38–145 neurons; 3.8k–35.9k training bins; 6-D behaviour |
| Encoder | CEBRA offset36-model-more-dropout, 64-D output, 64 hidden units, batch 2,048, 3,000 iterations, temperature 0.4, learning rate $3\times10^{-4}$; time-only fit(X) or behaviour-supervised fit(X, y) with time_delta conditional (hybrid = False) |
| Decoder | 2-layer MLP (64 hidden, LayerNorm, ReLU, dropout 0.4), Adam $10^{-3}$, 2,500 epochs, trained on training labels; metric: validation $R^2$ averaged over outputs |
| Training attack | 10 steps; raw-L2 PGD (step ε/5) or d′-APGD (1 restart, no warm-up or curriculum) |
| Evaluation | two shared metrics; decoder-targeted APGD, 20 steps, budgets 0.5–16; DDN margins (1,024 windows, 100 steps); equal-damage detectability (2,048 windows, 300 detector steps) |
| Replication | seeds 42, 43, 44; reproducible batch sampler (fixed in the fork); 4× H200 |

**Table 3.** Training arms. PGD6 (the tuned ACORN setting) is the primary baseline for all paired comparisons.

| Arm | Budget (training) | Geometry $\Sigma_s \otimes \Sigma_t$ | Residuals | Used in |
|---|---|---|---|---|
| CLEAN | — | — | — | all |
| PGD6 / PGD3 | raw L2, ε = 6 / 3 | isotropic, raw units (ACORN) | — | all / supervised |
| ISO8 | d′ = 8 | $\bar{\sigma}^2 I \otimes I$ ($\bar{\sigma}$-scaled L2 with APGD) | whole | all |
| SP8 | d′ = 8 | Ledoit–Wolf ⊗ I | whole | all |
| DIFF8 | d′ = 8 | Ledoit–Wolf ⊗ I | difference | all |
| POOL | d′ = 8 | {Ledoit–Wolf, $\bar{\sigma}^2 I$} ⊗ I, sampled per step | whole | all |
| REG8 | d′ = 8 | Ledoit–Wolf ⊗ I | regression on behaviour | supervised |
| NA4 | d′ = 4 | Ledoit–Wolf ⊗ Welch (full Kronecker) | whole | Exp. 5 |
| SP4 / SP16, ISO16, PGD12 | d′ 4/16, d′ 16, raw 12 | as above | whole | Exp. 5 sweep |
| AUTO | $\varepsilon_{\text{blind}}$ (8.3–12.7) | Ledoit–Wolf ⊗ I | whole | Exp. 5 |

# 5 Results

## 5.1 Equal nominal ε is not an equal perturbation

Every d′-trained run logs its realised perturbations in all units. At the same nominal budget d′ = 8, the realised raw L2 is 4.5 (ISO8), 5.2 (DIFF8), 5.5 (POOL) and 7.1 (SP8) spike counts (Figure 3b, Table 4). Once size is accounted for, clean accuracy hardly depends on geometry: all moderate budgets are within ±0.013 of PGD6 (e.g. SP8 +0.008, ISO8 +0.007, DIFF8 +0.012), and clean $R^2$ drops only when the realised perturbation exceeds ≈10 (PGD12 −0.085, SP16 −0.098; Figure 3a). Geometry changes *what the model becomes robust to*, not how well it decodes clean data.

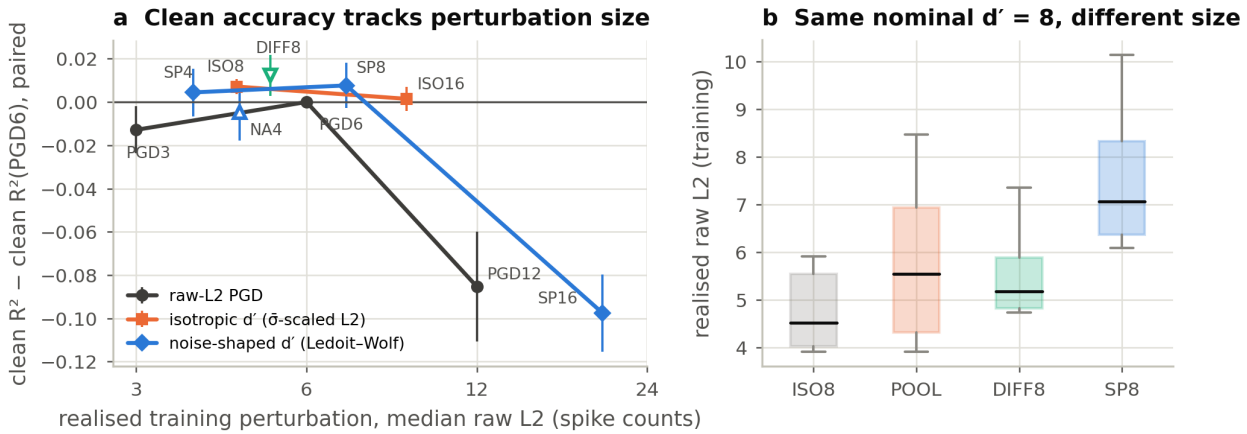



**Figure 3.** (a) Clean R² relative to PGD6 (paired within session, 9-session mean ± s.e., Exp. 5) against the realised training perturbation. (b) Realised raw L2 of the training perturbations of the four d′ = 8 arms (27 runs each, time-only).

**Table 4.** Realised training perturbations (medians over runs). Loss gain is the increase of the InfoNCE loss caused by the attack; values approaching ≈2.3 (SP16) indicate an attack that destroys the contrastive signal.

| CEBRA mode | Arm | raw L2 | reference d′ | isotropic d′ | loss gain | runs |
|---|---|---|---|---|---|---|
| time-only | SP4 | 3.78 | 3.39 | 6.13 | 0.38 | 9 |
| | NA4 | 4.57 | 4.00 | 7.35 | 0.43 | 9 |
| | ISO8 | 4.52 | 4.88 | 8.00 | 0.51 | 27 |
| | DIFF8 | 5.18 | 5.00 | 9.17 | 0.57 | 27 |
| | POOL | 5.55 | 6.94 | 8.00 | 0.80 | 27 |
| | SP8 | 7.06 | 6.94 | 12.00 | 0.76 | 27 |
| | ISO16 | 9.02 | 10.02 | 16.00 | 1.20 | 9 |
| | AUTO | 9.16 | 8.12 | 13.80 | 1.08 | 9 |
| | SP16 | 20.01 | 13.22 | 28.89 | 2.28 | 9 |
| supervised | ISO8 | 4.52 | 6.30 | 7.99 | 0.89 | 27 |
| | DIFF8 | 4.96 | 5.87 | 8.78 | 0.95 | 27 |
| | POOL | 4.52 | 7.43 | 8.00 | 1.03 | 27 |
| | REG8 | 6.39 | 6.83 | 11.30 | 1.13 | 27 |
| | SP8 | 7.65 | 7.25 | 12.15 | 1.20 | 27 |

## 5.2 Time-only CEBRA: noise-shaped robustness replicates in 27/27 runs

**Table 5.** Time-only CEBRA, paired with PGD6 within session and seed (9 sessions × 3 seeds). Ratios are geometric means over sessions with 90% t-intervals; 'wins' counts sessions (seed-averaged) and runs; p: exact sign test over sessions / t-test over the four animal means (noise-shaped threshold). POOL's clean-R² gain is 26/27 runs, 9/9 sessions, animal-level p = 0.007.

| Arm | Δ clean R² vs PGD6 | noise-shaped $\varepsilon_{1/2}$ ratio | wins (sessions · runs) | isotropic $\varepsilon_{1/2}$ ratio | wins | p (sessions / animals) |
|---|---|---|---|---|---|---|
| ISO8 | 0.000 [−0.003, +0.002] | 0.97× [0.97, 0.98] | 0/9 · 3/27 | 0.97× [0.95, 0.98] | 0/9 | 0.004 / 0.003 |
| SP8 | −0.002 [−0.030, +0.026] | 1.29× [1.14, 1.45] | 9/9 · 27/27 | 0.49× [0.36, 0.66] | 0/9 | 0.004 / 0.067 |
| DIFF8 | +0.010 [−0.005, +0.025] | 1.25× [1.11, 1.42] | 9/9 · 27/27 | 0.66× [0.55, 0.81] | 0/9 | 0.004 / 0.058 |
| POOL | +0.022 [+0.015, +0.030] | 1.21× [1.12, 1.31] | 9/9 · 27/27 | 0.82× [0.76, 0.88] | 0/9 | 0.004 / 0.024 |

All three noise-aware variants raise the noise-shaped threshold in every session and every seed (Figure 4), while the isotropic control ISO8 — the same APGD machinery with an isotropic metric — behaves like PGD6

(0.97×). The gain therefore comes from the geometry, not from the d′ parametrisation or the optimiser. Clean R² is unchanged for SP8 and DIFF8 (90% intervals within ±0.03) and *improved* for POOL (+0.022, 26/27 runs), which is also the most seed-stable arm (within-session s.d. 0.007 vs 0.012 for PGD6). The cost is isotropic robustness: 0.49× (SP8), 0.66× (DIFF8) and 0.82× (POOL); the full robustness curves are in Figure 5.

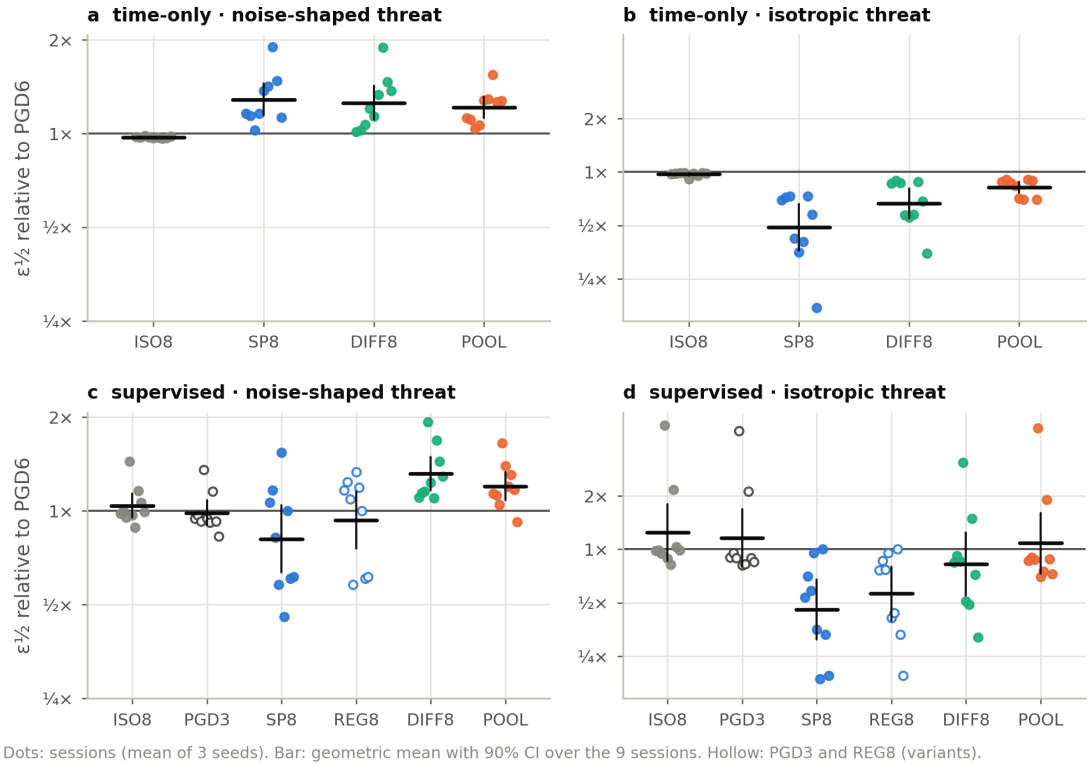



**Figure 4.** Half-damage thresholds relative to PGD6 for each session (dots, mean of 3 seeds) with the geometric mean and 90% interval (bar), under the noise-shaped (left) and isotropic (right) attack, for time-only (top) and behaviour-supervised (bottom) CEBRA.

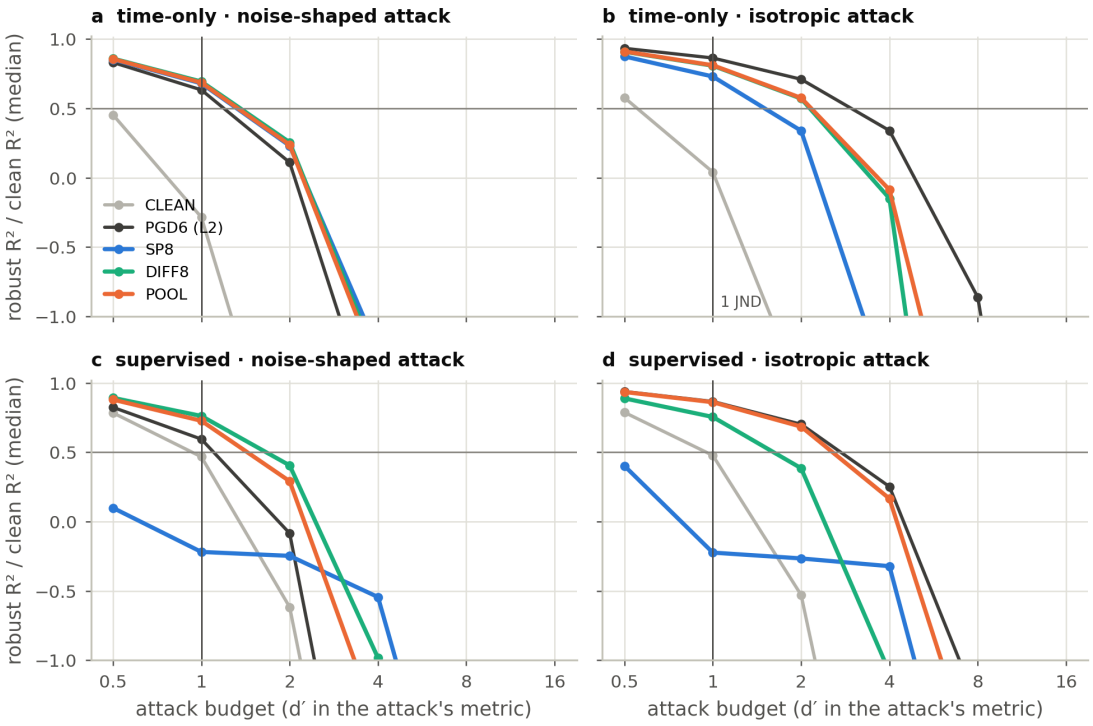



**Figure 5.** Median robust R² relative to clean R² against the attack budget (27 runs per curve). Noise-aware arms lie above PGD6 under the noise-shaped attack and below it under the isotropic attack; in supervised CEBRA the whole-covariance SP8 collapses already at $d' = 0.5$ (panel c), while DIFF8 and POOL do not.

## 5.3 Behaviour-supervised CEBRA: the noise model must describe noise

**Table 6.** Behaviour-supervised CEBRA (Exp. 8), paired with PGD6; conventions as in Table 5. PGD3 is ACORN's supervised default.

| Arm | Δ clean R² vs PGD6 | noise-shaped $\varepsilon_{1/2}$ ratio | wins (sessions · runs) | noise-shaped ratio vs PGD3 | isotropic $\varepsilon_{1/2}$ ratio | p (sessions / animals) |
|---|---|---|---|---|---|---|
| PGD3 | +0.007 [−0.006, +0.019] | 0.99× [0.90, 1.08] | 2/9 · 6/27 | — | 1.16× [0.80, 1.68] | 0.180 / 0.421 |
| ISO8 | +0.005 [−0.008, +0.019] | 1.04× [0.95, 1.13] | 3/9 · 12/27 | 1.05× | 1.25× [0.86, 1.80] | 0.508 / 0.614 |
| SP8 | −0.001 [−0.009, +0.006] | 0.81× [0.64, 1.04] | 3/9 · 9/27 | 0.83× | 0.46× [0.31, 0.68] | 0.508 / 0.046 |
| REG8 | 0.000 [−0.006, +0.006] | 0.93× [0.76, 1.15] | 5/9 · 15/27 | 0.95× | 0.56× [0.40, 0.80] | 1.000 / 0.452 |
| DIFF8 | +0.011 [−0.005, +0.026] | 1.32× [1.16, 1.49] | 9/9 · 27/27 | 1.34× | 0.82× [0.55, 1.24] | 0.004 / 0.020 |
| POOL | +0.011 [−0.001, +0.022] | 1.20× [1.08, 1.33] | 8/9 · 25/27 | 1.22× | 1.08× [0.73, 1.60] | 0.039 / 0.157 |

Supervision raises clean R² for every arm (+0.24 for CLEAN, +0.08–0.10 for the AT arms; 27/27 runs) and removes AT's clean-accuracy benefit: all arms are within ±0.011 of PGD6. The noise-shaped robustness gain survives for **DIFF8** (1.32× vs PGD6 in 9/9 sessions and 27/27 runs, animal-level p = 0.02; 1.34× vs PGD3) and **POOL** (1.20×, 8/9 sessions; 1.22× vs PGD3), and in this mode POOL has no measurable isotropic cost (1.08×, n.s.).

The whole-covariance **SP8** and the regression-residual **REG8** do not: in 5/9 sessions their decoders lose half of R² already at d′ = 0.5 (Figure 5c) and under a single draw of natural noise (natural-noise R² 0.09 and −0.07, against 0.38 for PGD6 and 0.53 for DIFF8). Both have the largest realised perturbations (raw L2 7.6 and 6.4 vs 4.5–5.0) and loss gains (1.20 and 1.13 vs 0.89–1.04): a supervised objective is easier to attack, so the same nominal d′ over-shoots. Linear regression on six behavioural variables removes little of the slow, high-variance population signal; the difference residual removes it and yields a smaller, safer metric. This is the practical lesson of the supervised experiment: estimate Σ on the noise, not on the signal.

## 5.4 The trade-off map

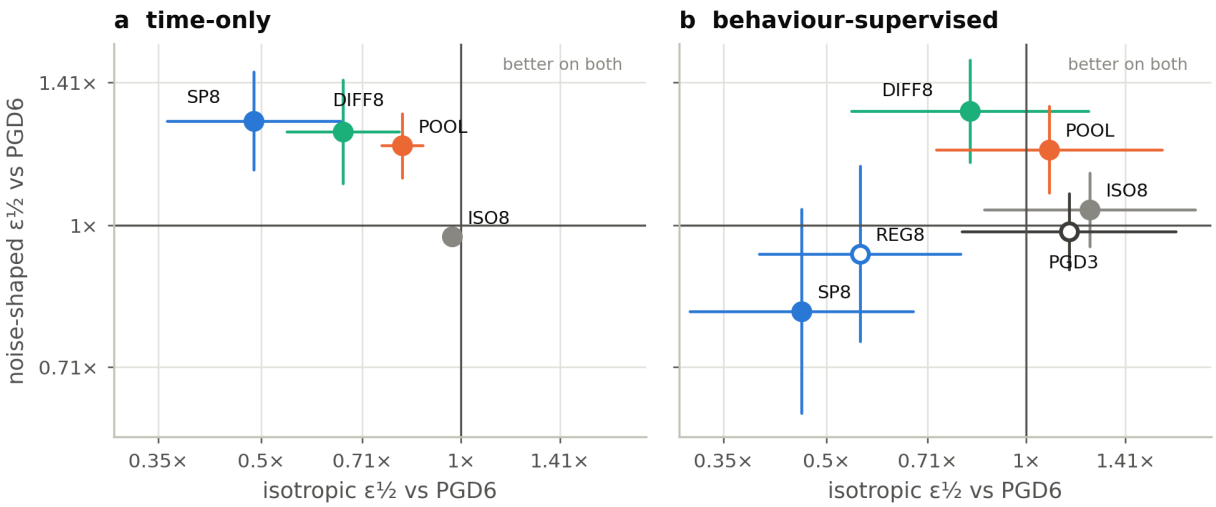



**Figure 6.** Geometric-mean ratios to PGD6 of the noise-shaped (vertical) and isotropic (horizontal) half-damage thresholds, with 90% intervals over sessions. Upper-right is better on both threats. POOL moves towards that quadrant in time-only CEBRA and reaches it in supervised CEBRA.

Figure 6 places every arm on both threat axes at once. In time-only CEBRA the three noise-aware arms lie on a frontier above PGD6 on the noise-shaped axis and below it on the isotropic one: SP8 buys the largest gain with the largest cost (1.29×, 0.49×), DIFF8 sits in between (1.25×, 0.66×) and POOL gives up the least (1.21×, 0.82×). ISO8 coincides with PGD6 (0.97×, 0.97×), so the d′ machinery alone changes nothing. In supervised CEBRA, DIFF8 keeps its time-only position (1.32×, 0.82×), POOL moves into the upper-right quadrant (1.20×, 1.08×; the isotropic interval includes 1), and SP8 and REG8 fall below PGD6 on both axes.

## 5.5 Detectability at equal damage

**Table 7.** Detector AUC and realised reference d′ of the attacks that halve each model's $R^2$ (9-session means, saved seed-42 time-only models). The blind autoencoder (AE) never sees an attack; the CNN is trained on clean versus attacked windows of the same model and geometry.

| Model | Noise-shaped attack | | | Isotropic attack | | |
|---|---|---|---|---|---|---|
| | AE AUC | CNN AUC | ref. d′ | AE AUC | CNN AUC | ref. d′ |
| CLEAN | 0.501 | 0.534 | 0.55 | 0.536 | 0.798 | 1.68 |
| PGD6 | 0.502 | 0.570 | 1.41 | 0.503 | 0.551 | 2.62 |
| ISO8 | 0.502 | 0.570 | 1.38 | 0.503 | 0.556 | 2.66 |
| SP8 | 0.501 | 0.575 | 1.75 | 0.537 | 0.778 | 4.46 |
| DIFF8 | 0.501 | 0.575 | 1.68 | 0.562 | 0.793 | 5.53 |
| NA4 | 0.501 | 0.565 | 1.83 | 0.529 | 0.767 | 4.35 |

Training geometry also changes *which* attacks remain effective. For L2-trained decoders the two geometries are equally stealthy at equal damage (CNN AUC 0.57 vs 0.55). For noise-aware decoders the isotropic attacks that still halve $R^2$ must move along low-variance, unnatural directions (realised reference d′ 4.5 vs 2.6) and are readily detected in the low-firing sessions (CNN AUC up to 0.98; Spearman $\rho = -0.70$ for SP8 and $-0.75$ for DIFF8 between the detectability gap and the session's median noise SD; Figure 7). Measured in d′, the cheapest attack that halves $R^2$ is the noise-shaped one for every model in every session, and noise-aware training raises its cost 1.26–1.37× (9/9 sessions). Honest caveat: in units of the supervised CNN's AUC, the stealthiest half-damage attack is about as stealthy for PGD6 (0.55) as for SP8 (0.57) — the gain is a gain in required budget, not in stealth.

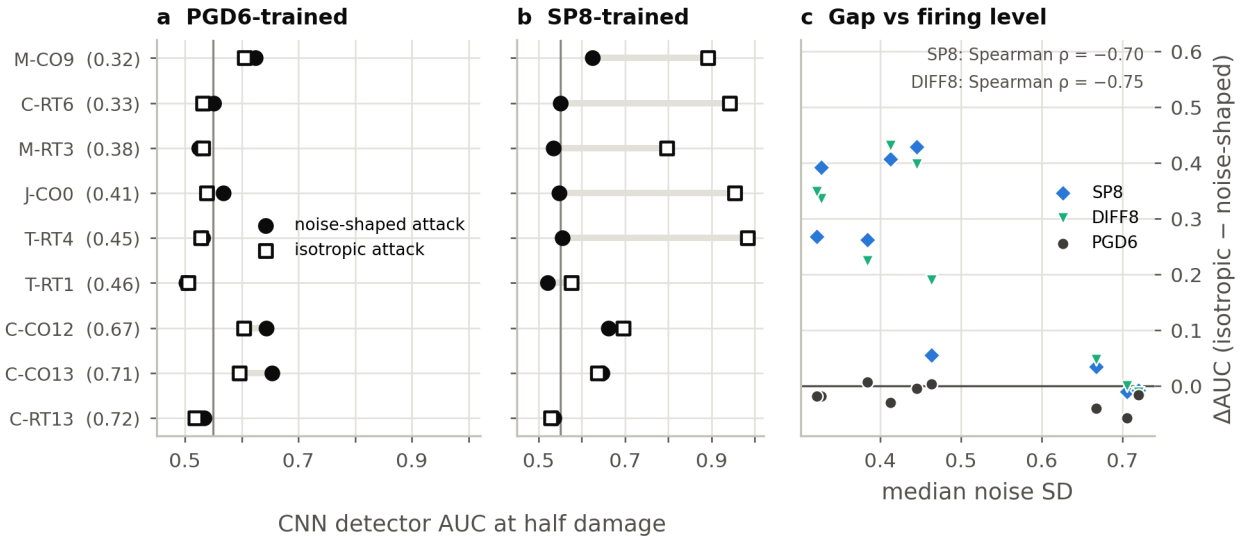



**Figure 7.** (a, b) CNN detector AUC of the noise-shaped (filled) and isotropic (open) attacks at half damage, for PGD6- and SP8-trained decoders; sessions ordered by median noise SD (in parentheses), the vertical line is the detection criterion AUC = 0.55 of Section 3.5. (c) Detectability gap (isotropic minus noise-shaped CNN AUC) against the session's median noise SD for PGD6-, SP8- and DIFF8-trained decoders.

## 5.6 Robustness margin relative to the JND

The note proposed the margin $\varepsilon_{1/2}/\varepsilon_{JND}$ as a dataset-level summary. In time-only CEBRA no clean decoder clears one JND (0/27), PGD6 and ISO8 clear it in 18/27 runs, SP8 in 25/27, POOL in 21/27 and DIFF8 in **27/27** — DIFF8 is also the only arm that clears it in both metrics in every run. In supervised CEBRA the clean decoders are more robust (12/27), PGD3/PGD6 reach 17/27 and 15/27, and DIFF8 again leads (24/27), followed by POOL (20/27); SP8 drops to 11/27 (Figure 8).

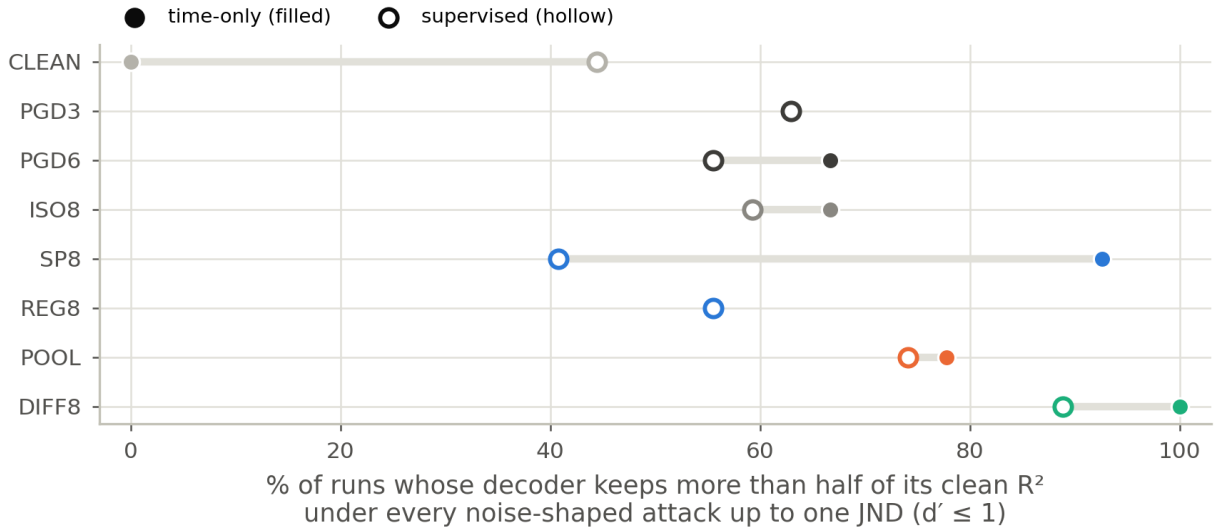



**Figure 8.** Share of runs whose decoder keeps more than half of its clean R² against every noise-shaped perturbation up to one JND, in time-only (filled) and supervised (open) CEBRA.

## 5.7 Can ε be chosen automatically?

**Table 8.** Data-defined budgets per session (Exp. 5). $\varepsilon_{blind}$: blind-detector crossing of AUC 0.55 (log-interpolated); $\varepsilon_{det}$: largest grid value at which both detectors stay ≤ 0.55.

| Session | C | D = C·T | $\sqrt{(2D)}$ | $\varepsilon_{NN}$ | $(2D)^{1/4}$ | $\varepsilon_{blind}$ (AE) | $\varepsilon_{det}$ | $\kappa(\Sigma_s)$ | median SD |
|---|---|---|---|---|---|---|---|---|---|
| C-CO12 | 86 | 3096 | 78.7 | 69.8 | 8.87 | 8.74 | 4 | 55 | 0.67 |
| C-CO13 | 72 | 2592 | 72.0 | 63.4 | 8.49 | 9.03 | 4 | 22 | 0.71 |
| C-RT13 | 75 | 2700 | 73.5 | 64.4 | 8.57 | 8.30 | 2 | 43 | 0.72 |
| C-RT6 | 63 | 2268 | 67.3 | 51.4 | 8.21 | 8.85 | 1 | 546 | 0.33 |
| M-CO9 | 145 | 5220 | 102.2 | 85.6 | 10.11 | 10.79 | 4 | 1202 | 0.32 |
| M-RT3 | 116 | 4176 | 91.4 | 73.9 | 9.56 | 12.72 | 2 | 2272 | 0.38 |
| T-RT1 | 64 | 2304 | 67.9 | 57.6 | 8.24 | 9.48 | 2 | 484 | 0.46 |
| T-RT4 | 53 | 1908 | 61.8 | 50.8 | 7.86 | 9.24 | 2 | 1493 | 0.45 |
| J-CO0 | 38 | 1368 | 52.3 | 39.1 | 7.23 | 9.73 | 2 | 4905 | 0.41 |

The calibration is internally consistent: the theoretical blind threshold $(2D)^{1/4}$ predicts the measured one within 13% on average, and the learned detectors agree with the ideal-observer JND at small d′. As a *training* budget, however, the blind threshold is 1.2–3× larger than the per-session optimum (d′ = 4 in five sessions and 8 in four): AUTO matches PGD6 on average (clean +0.003) but over-shoots in M-RT3 and J-CO0, where the training loss gain saturates (≈ 2.3) as for SP16. We therefore treat $\varepsilon_{JND}$ as the evaluation threat budget and a fixed d′ (or the pool) as the training budget.

# 6 Discussion and next steps

**What is established.** (i) A detectability metric gives ε a meaning that is shared across recordings: detector curves, JND and natural-variability scales line up across nine sessions and four animals. (ii) Under that metric, current neural decoders — clean or L2-trained — are vulnerable within or just beyond one JND. (iii) Noise-aware AT raises the noise-shaped robustness threshold by 1.2–1.3× reproducibly (27/27 time-only runs; 27/27 and 25/27 for DIFF8 and POOL in supervised CEBRA) without cost in clean decoding. (iv) The trade-off with isotropic robustness is real but structured: it is smallest for POOL and, in low-firing sessions, concerns attacks that are easy to detect.

**Recommended configuration.** DIFF8 as the principled default — label-free, consistent in both CEBRA modes, the best JND margin (27/27 and 24/27); POOL when clean accuracy and isotropic robustness matter (the only arm with a consistent clean gain, +0.022, 9/9 sessions). The whole-covariance SP8 gives the largest gain in time-only CEBRA but is unsafe with a supervised encoder.

**Limitations.** One dataset family (Perich; animal-level p-values between 0.007 and 0.07). Perturbations are continuous offsets on integer counts, not spike insertions or deletions. The reference threat model is itself an estimate (nearest-Kronecker error 0.34–0.66 against a sampling floor of 0.18–0.35). Robustness is measured with white-box APGD and DDN — monotone worst-case curves and agreement between the two attack families argue against gradient masking, but a multi-restart stress test is still pending. The automatic training budget is not decoder-optimal.

**Next steps.** (1) External validation on independent animals (e.g. the rat-hippocampus and monkey area-2 datasets used by CEBRA) with the frozen protocol. (2) A strong-attack sanity check (more steps and restarts) on the saved checkpoints. (3) Writing the paper around the three contributions: the noise-calibrated threat model and its calibration, the vulnerability of current decoders, and noise-aware training. Positioning: covariance-shaped perturbation sets exist for tabular data (Erdemir et al., NeurIPS 2021) and detectability-bounded attacks for reinforcement learning (Franzmeyer et al., ICLR 2024); to our knowledge, neither a noise-calibrated threat model for neural data nor adversarial robustness of CEBRA embeddings has been studied.

# Appendix A Per-session results (mean of 3 seeds)

**Table 9.** Time-only CEBRA — clean validation $R^2$. Bold: best arm of the session.

| Session | CLEAN | PGD6 | ISO8 | SP8 | DIFF8 | POOL |
|---|---|---|---|---|---|---|
| C-CO12 | 0.466 | 0.626 | 0.628 | 0.638 | 0.614 | **0.640** |
| C-CO13 | 0.507 | 0.713 | 0.713 | 0.711 | 0.691 | **0.722** |
| C-RT13 | 0.308 | 0.421 | 0.427 | 0.431 | 0.427 | **0.440** |
| C-RT6 | 0.362 | 0.542 | 0.538 | 0.531 | 0.547 | **0.554** |
| M-CO9 | 0.512 | 0.700 | 0.694 | 0.694 | 0.696 | **0.717** |
| M-RT3 | 0.363 | 0.499 | 0.502 | 0.516 | 0.521 | **0.538** |
| T-RT1 | 0.237 | 0.314 | 0.315 | 0.375 | **0.375** | 0.359 |
| T-RT4 | 0.160 | 0.187 | 0.183 | 0.195 | 0.194 | **0.203** |
| J-CO0 | 0.328 | 0.560 | 0.561 | 0.452 | 0.586 | **0.588** |

**Table 10.** Time-only CEBRA — noise-shaped $\varepsilon_{1/2}$ (values of 0.50 are clipped at the smallest budget). Bold: best arm of the session.

| Session | CLEAN | PGD6 | ISO8 | SP8 | DIFF8 | POOL |
|---|---|---|---|---|---|---|
| C-CO12 | 0.66 | 2.27 | 2.21 | **2.63** | 2.29 | 2.55 |
| C-CO13 | 0.68 | 2.28 | 2.22 | **2.60** | 2.34 | 2.53 |
| C-RT13 | 0.57 | 1.28 | 1.27 | 1.32 | **1.37** | 1.33 |
| C-RT6 | 0.50 | 1.09 | 1.06 | 1.27 | **1.31** | 1.16 |
| M-CO9 | 0.55 | 2.14 | 2.07 | **2.93** | 2.43 | 2.74 |
| M-RT3 | 0.50 | 1.21 | 1.17 | **1.71** | 1.61 | 1.56 |
| T-RT1 | 0.50 | 0.56 | 0.54 | **1.06** | 1.06 | 0.87 |
| T-RT4 | 0.50 | 0.73 | 0.71 | **1.08** | 1.07 | 0.92 |
| J-CO0 | 0.50 | 0.82 | 0.80 | 0.93 | **1.12** | 1.04 |

**Table 11.** Time-only CEBRA — isotropic $\varepsilon_{1/2}$. Bold: best arm of the session.

| Session | CLEAN | PGD6 | ISO8 | SP8 | DIFF8 | POOL |
|---|---|---|---|---|---|---|
| C-CO12 | 1.11 | **4.90** | 4.78 | 3.39 | 4.20 | 4.32 |
| C-CO13 | 1.13 | **4.55** | 4.44 | 3.28 | 4.04 | 4.11 |
| C-RT13 | 0.91 | **2.50** | 2.46 | 1.82 | 2.16 | 2.16 |
| C-RT6 | 0.50 | **2.50** | 2.46 | 1.05 | 1.43 | 2.09 |
| M-CO9 | 0.68 | **6.47** | 5.91 | 2.28 | 3.60 | 4.58 |
| M-RT3 | 0.54 | **3.85** | 3.75 | 1.55 | 2.22 | 2.69 |
| T-RT1 | 0.50 | **1.43** | 1.36 | 1.04 | 1.25 | 1.28 |
| T-RT4 | 0.50 | **1.57** | 1.54 | 0.90 | 1.07 | 1.40 |
| J-CO0 | 0.50 | **2.95** | 2.88 | 0.51 | 1.02 | 2.05 |

**Table 12.** Behaviour-supervised CEBRA — clean validation $R^2$. Bold: best arm of the session.

| Session | CLEAN | PGD3 | PGD6 | ISO8 | SP8 | REG8 | DIFF8 | POOL |
|---|---|---|---|---|---|---|---|---|
| C-CO12 | 0.683 | 0.682 | 0.679 | 0.683 | 0.686 | 0.679 | 0.683 | **0.696** |
| C-CO13 | 0.777 | 0.775 | 0.771 | 0.769 | 0.759 | 0.780 | **0.784** | 0.769 |
| C-RT13 | 0.537 | 0.540 | 0.537 | 0.538 | 0.539 | 0.546 | **0.546** | 0.545 |
| C-RT6 | 0.647 | 0.645 | 0.644 | 0.661 | 0.649 | 0.660 | **0.665** | 0.660 |
| M-CO9 | 0.772 | 0.778 | 0.785 | 0.772 | 0.783 | **0.788** | 0.785 | 0.782 |
| M-RT3 | 0.581 | 0.595 | 0.602 | 0.583 | 0.583 | 0.597 | 0.598 | **0.610** |
| T-RT1 | **0.484** | 0.461 | 0.401 | 0.457 | 0.421 | 0.387 | 0.461 | 0.455 |
| T-RT4 | 0.250 | 0.252 | 0.252 | 0.263 | 0.248 | 0.249 | **0.278** | 0.264 |
| J-CO0 | 0.685 | **0.706** | 0.701 | 0.694 | 0.693 | 0.686 | 0.667 | 0.687 |

**Table 13.** Behaviour-supervised CEBRA — noise-shaped $\varepsilon_{1/2}$ (values of 0.50 are clipped at the smallest budget). Bold: best arm of the session.

| Session | CLEAN | PGD3 | PGD6 | ISO8 | SP8 | REG8 | DIFF8 | POOL |
|---|---|---|---|---|---|---|---|---|
| C-CO12 | 1.41 | 2.27 | 2.41 | 2.36 | 2.57 | **2.80** | 2.66 | 2.74 |
| C-CO13 | 1.50 | 2.29 | 2.36 | 2.36 | 2.76 | **2.92** | 2.69 | 2.65 |
| C-RT13 | 1.14 | 1.60 | 1.73 | 1.65 | 1.50 | 1.88 | **2.00** | 1.81 |
| C-RT6 | 0.93 | 1.17 | 0.86 | 1.24 | 0.50 | 0.50 | **1.66** | 1.42 |
| M-CO9 | 1.18 | 2.16 | 2.29 | 2.21 | **3.53** | 3.05 | 2.82 | 3.20 |
| M-RT3 | 0.60 | 1.00 | 1.09 | 0.97 | 0.50 | 1.30 | 1.20 | **1.31** |
| T-RT1 | 0.50 | 0.58 | 0.50 | 0.58 | 0.50 | 0.50 | **0.85** | 0.65 |
| T-RT4 | 0.52 | 0.76 | 0.83 | 0.88 | 0.50 | 0.50 | **1.19** | 0.96 |
| J-CO0 | 0.65 | 0.67 | 0.81 | 0.81 | 0.50 | 0.50 | **1.05** | 0.78 |

**Table 14.** Behaviour-supervised CEBRA — isotropic $\varepsilon_{1/2}$. Bold: best arm of the session.

| Session | CLEAN | PGD3 | PGD6 | ISO8 | SP8 | REG8 | DIFF8 | POOL |
|---|---|---|---|---|---|---|---|---|
| C-CO12 | 2.17 | 4.54 | **5.06** | 4.95 | 2.73 | 3.86 | 4.27 | 4.36 |
| C-CO13 | 2.14 | 4.34 | **4.57** | 4.52 | 3.25 | 3.93 | 4.21 | 4.11 |
| C-RT13 | 1.55 | 2.67 | **3.00** | 2.83 | 1.84 | 2.30 | 2.56 | 2.62 |
| C-RT6 | 0.96 | 2.43 | 0.53 | **2.61** | 0.50 | 0.50 | 1.61 | 2.52 |
| M-CO9 | 1.34 | 5.45 | **6.71** | 5.92 | 2.37 | 2.75 | 3.42 | 4.69 |
| M-RT3 | 0.67 | 2.23 | **2.69** | 2.21 | 0.50 | 1.18 | 1.31 | 2.02 |
| T-RT1 | 0.50 | 1.06 | 0.50 | **1.08** | 0.50 | 0.50 | 0.75 | 0.95 |
| T-RT4 | 0.51 | 1.35 | 1.51 | **1.55** | 0.50 | 0.50 | 1.08 | 1.32 |
| J-CO0 | 0.77 | 2.18 | **2.56** | 2.52 | 0.50 | 0.50 | 0.82 | 2.02 |

# Appendix B Exp. 5 sweep (time-only, seed 42, 9-session means)

**Table 15.** All twelve arms of the nine-session sweep. Raw L2 of PGD arms is their ε (the projection is active).

| Arm | clean $R^2$ | Δ vs PGD6 | noise-shaped $\varepsilon_{1/2}$ | isotropic $\varepsilon_{1/2}$ | raw L2 (training) | ref. d′ (training) | loss gain |
|---|---|---|---|---|---|---|---|
| CLEAN | 0.365 | −0.137 | 0.56 | 0.72 | — | — | — |
| PGD3 | 0.488 | −0.013 | 1.29 | 3.16 | 3.00 | — | — |
| PGD6 | 0.501 | (base) | 1.37 | 3.36 | 6.00 | — | — |
| PGD12 | 0.416 | −0.085 | 0.79 | 1.17 | 12.00 | — | — |
| ISO8 | 0.508 | +0.007 | 1.35 | 3.31 | 4.52 | 4.93 | 0.51 |
| ISO16 | 0.502 | +0.001 | 1.19 | 2.67 | 9.02 | 10.02 | 1.20 |
| SP4 | 0.505 | +0.004 | 1.65 | 2.01 | 3.78 | 3.39 | 0.38 |
| SP8 | 0.509 | +0.008 | 1.71 | 1.77 | 7.06 | 6.94 | 0.76 |
| SP16 | 0.403 | −0.098 | 0.57 | 0.76 | 20.01 | 13.22 | 2.28 |
| NA4 | 0.496 | −0.005 | 1.77 | 1.82 | 4.57 | 4.00 | 0.43 |
| DIFF8 | 0.513 | +0.012 | 1.62 | 2.31 | 5.18 | 5.01 | 0.57 |
| AUTO | 0.504 | +0.003 | 1.56 | 1.59 | 9.16 | 8.12 | 1.08 |

# Appendix C Noise-model diagnostics per session

**Table 16.** Relative Frobenius errors of a Ledoit–Wolf ⊗ white model and of the nearest Kronecker product against the empirical full covariance of 8 channels × 36 bins, with the sampling floor of that estimate.

| Session | neurons | training bins | median noise SD | $\kappa(\Sigma_s)$ | error of LW ⊗ I | error of nearest Kronecker | sampling floor |
|---|---|---|---|---|---|---|---|
| C-CO12 | 86 | 18384 | 0.668 | 55 | 0.85 | 0.56 | 0.18 |
| C-CO13 | 72 | 17861 | 0.706 | 22 | 0.84 | 0.54 | 0.19 |
| C-RT13 | 75 | 35938 | 0.720 | 43 | 0.72 | 0.50 | 0.18 |
| C-RT6 | 63 | 8389 | 0.327 | 546 | 0.75 | 0.45 | 0.35 |
| M-CO9 | 145 | 13604 | 0.322 | 1202 | 0.89 | 0.66 | 0.18 |
| M-RT3 | 116 | 9174 | 0.384 | 2272 | 0.86 | 0.34 | 0.22 |
| T-RT1 | 64 | 11458 | 0.464 | 484 | 0.70 | 0.37 | 0.25 |
| T-RT4 | 53 | 10360 | 0.445 | 1493 | 0.71 | 0.47 | 0.31 |
| J-CO0 | 38 | 3816 | 0.413 | 4905 | 0.89 | 0.52 | 0.21 |